Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 12)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows features of three rechargeable battery types.

Table 1

Battery type	Energy density / Wh kg^{-1}	Charge-discharge cycles	Storage efficiency / %	Cost per kWh / £	Safety issues
Lead-acid	35	500	80	100	Toxic lead, acid
Nickel-cadmium	50	1500	75	250	Toxic cadmium
Lithium-ion	200	2000	92	150	Fire risk

0 1 . 1 Use Table 1 to explain why lithium-ion batteries are used in electric vehicles but lead-acid batteries are still used for stationary storage in some off-grid solar systems.

[4 marks]

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0 1 . 2 Give **two** other factors that affect the viability of a rechargeable battery.

[2 marks]

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0 1 . 3 Describe how Vehicle-to-Grid (V2G) systems could reduce the need for peaking power stations.

[2 marks]

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0 1 . 4 Explain how Power-to-Gas (P2G) systems use existing infrastructure.

[2 marks]

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0 2 Figure 1 shows a simplified nitrogen cycle with some processes labelled 1 to 6.

Figure 1

Number	Process description
1	Nitrogen gas converted to ammonium ions by Rhizobium in root nodules
2	Ammonium ions oxidised to nitrite then nitrate by Nitrosomonas and Nitrobacter
3	Nitrates reduced to nitrogen gas by Pseudomonas in anaerobic soil
4	Amino groups in dead organic matter released as ammonium ions by decomposers
5	Nitrogen and oxygen react during lightning to form NOx
6	Nitrates washed out of soil into rivers

0 2 . 1 Name processes 1 to 6.

[3 marks]

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0 2 . 2 Explain why draining a waterlogged field increases the nitrate available to crops.

[3 marks]

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0 2 . 3 Explain why nitrogen is described as a resource that is 'not very abundant' for living organisms even though the atmosphere is 78% nitrogen.

[2 marks]

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0 2 . 4 Explain why growing legumes such as clover in a crop rotation reduces the need for artificial fertiliser.

[2 marks]

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0 3 A student investigated whether the number of earthworms per m^2 changes with distance from a hedge across an arable field.

0 3 . 1 Explain why systematic sampling along a transect is more appropriate than random sampling for this investigation.

[2 marks]

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0 3 . 2 Describe how a preliminary study could be used to decide the distance between sampling sites and the number of samples at each site.

[4 marks]

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0 3 . 3 Describe how earthworms could be extracted from a 0.5 m x 0.5 m sample area and give **two** limitations of the method.

[4 marks]

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0 4 Figure 1 lists the stages in a water treatment works supplying drinking water from a lowland river.

Figure 1

Stage	Process
1	Screens
2	Sedimentation
3	Aeration
4	Flocculation and clarification
5	Filtration through sand and gravel
6	Activated carbon
7	Sterilisation
8	pH adjustment

0 4 . 1 Explain why clay particles do not settle out in the sedimentation stage, and describe how they are removed in stage 4.

[3 marks]

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0 4 . 2 Explain the purpose of aeration (stage 3).

[2 marks]

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0 4 . 3 Explain why ozone or UV light may be used for sterilisation instead of chlorine when water comes from peaty upland sources.

[2 marks]

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0 4 . 4 Describe how each of the following contaminants is removed from water.

[3 marks]

Toxic metal ions (lead, mercury, arsenic)

Salt

Odours and pesticides

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0 5 . 1 Define the term **persistence** as applied to a pollutant.

[1 mark]

0 5 . 2 Explain how the persistence of a pesticide is affected by **two** named environmental factors.

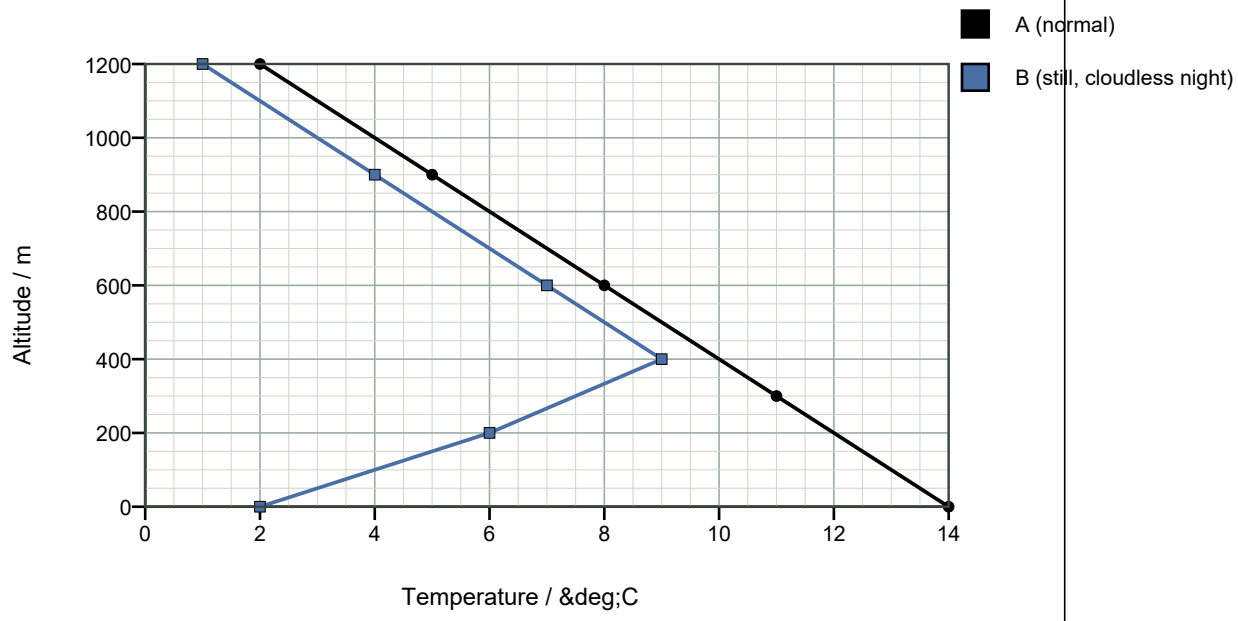
[4 marks]

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0 6 Figure 1 shows how air temperature changes with altitude above a city in a valley on two different nights. Line A shows normal conditions. Line B shows the conditions on a still, cloudless night.

Figure 1



0 6 . 1 Name the atmospheric condition shown by line B in Figure 1.

[1 mark]

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0 6 . 2 Use Figure 1 to explain why pollutants released at ground level would disperse on night A but not on night B.

[4 marks]

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0 6 . 3 Explain why the conditions on night B (still, cloudless) allowed an inversion to form.

[3 marks]

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0 6 . 4 Give **two** types of pollution event that are made more severe by temperature inversions.

[2 marks]

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0 7 . 1 Describe coal gasification and coal liquefaction and explain how each increases the usefulness of coal.

[4 marks]

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0 7 . 2 Describe what methane hydrate is and where it is found.

[2 marks]

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0 7 . 3 Describe **two** proposed methods of extracting methane from hydrate deposits and explain **two** environmental risks.

[4 marks]

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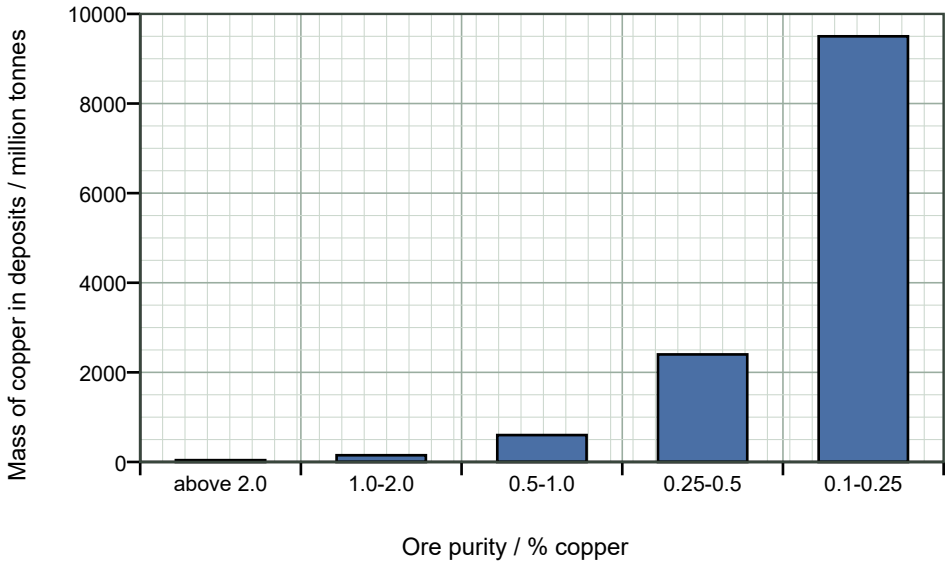
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0 8 Figure 1 shows the mass of copper contained in the Earth's crust at different ore purities, illustrating Lasky's principle.

Figure 1



0 8 . 1 State Lasky's principle.

[1 mark]

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0 8 . 2 Use Figure 1 to calculate how many times more copper is contained in deposits of 0.25-0.5% purity than in deposits above 2.0%.

[1 mark]

Answer ..... times

0 8 . 3 Explain the difference between the **reserves** and the **resource** of a mineral.

[3 marks]

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0 8 . 4 Explain, using Lasky's principle, why the ability to exploit slightly lower grade ores greatly increases reserves.

[2 marks]

0 8 . 5 Explain why the main problem for future mineral supplies is not the quantity that exists in the crust.

[3 marks]

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0 9 Table 1 shows the pH of rainwater and the number of fish species in lakes in four areas of Scandinavia with different bedrock geology.

Table 1

Lake area	Bedrock	Mean rainfall pH	Mean lake pH	Number of fish species	Dissolved aluminium / mg dm^{-3}
A	Granite	4.3	4.6	0	0.42
B	Granite	4.8	5.1	3	0.20
C	Limestone	4.3	7.2	8	0.01
D	Limestone	5.5	7.6	9	0.01

0 9 . 1 State the pH below which rain is described as acid rain and explain why unpolluted rain is already slightly acidic.

[2 marks]

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0 9 . 2 Use Table 1 to explain why lakes in area C contain many more fish species than lakes in area A, even though the rainfall has the same pH.

[4 marks]

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0 9 . 3 Describe **two** effects of acid rain on non-living objects.

[2 marks]

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0 9 . 4 Explain how sulfur dioxide released from a power station is converted to sulfuric acid in the atmosphere.

[2 marks]

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1 0 . 1 Evaluate the potential contribution of renewable energy resources to future energy supplies in a country such as the UK.

[9 marks]

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Extra space

1 0 . 2 State the factor by which the kinetic energy of wind increases when wind velocity doubles.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.

[25 marks]

OR

1 1 . 2 Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS